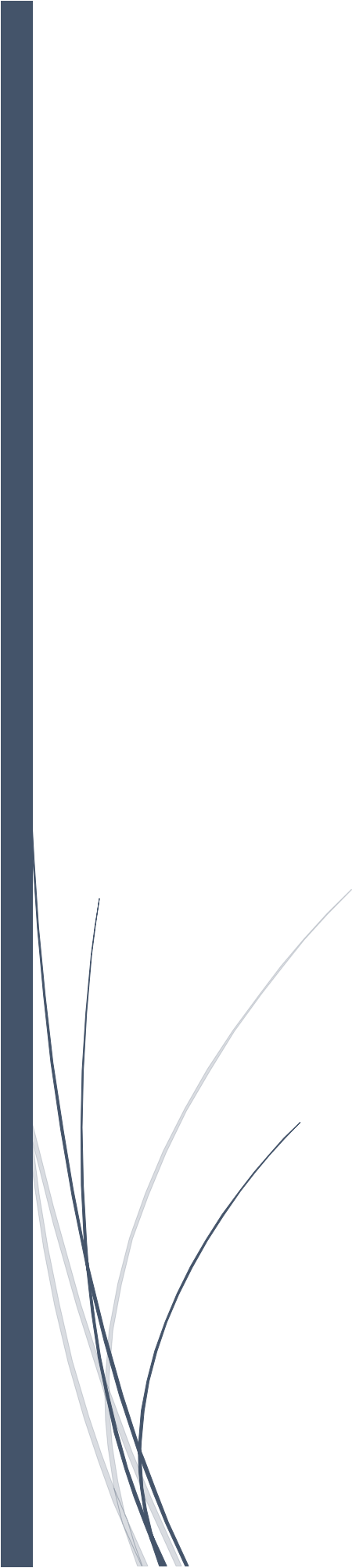


A thick dark blue vertical bar is positioned on the left side of the page. From its base, several thin, curved lines in shades of blue and grey extend upwards and outwards, creating an abstract, organic shape.

Orientation of switching inductor for EMI/EMC Compliance

Best Practice for EMC/EMI
Compliance series

Abdallah Ebraheem Abdelmotalb Keshk

For further Rule Of thumbs for reducing EMI

Orientation of the SWITCHING Inductor matters

First you have to know there is two kind of radiations when combined we say electromagnetic radiation

- 1- Magnetic Radiation
- 2- Electric Radiation

these radiations are perpendicular on each other in the Far-Field

In the Near-Field Measurements (Pre-Compliance Measurements) you have to use special antenna for each type of radiation H-Probe is used for Magnetic radiation and E-Probe is used for Electric Radiation

The problem of the Magnetic Radiated Emissions generated from the switching core is from the leakage flux of the inductor spreading out around the board as shown here

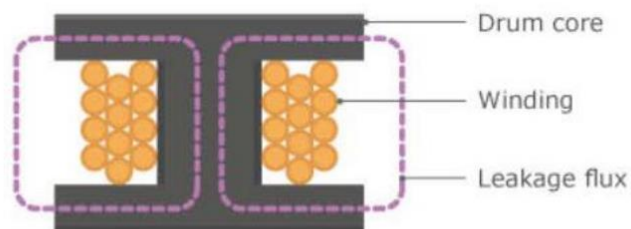


FIGURE 1: MAGNETIC FLUX IN UNSHIELDED DRUM CORE COILS

To mitigate this problem, you have to choose shielded inductor where the inductor core is surrounding the magnetic flux leaving no room for the leakage flux to escape ideally like below

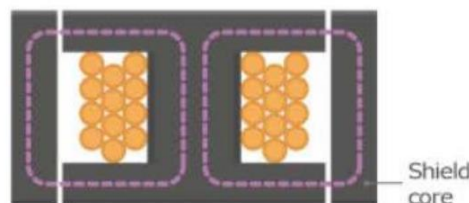


FIGURE 2: MAGNETIC FLUX IN SHIELDED DRUM CORE COILS

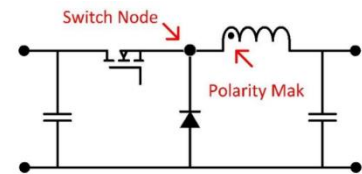


FIGURE 24: POWER STAGE OF DC-DC CONVERTER WITH DEFINED COIL ORIENTATION

But as the nature says there is no way to be an ideal component that means it will suffer from small amount of flux leakage to be pickup by the antenna but it's a huge improvement compared to the non-shielded inductor

Back to our point of the orientation and how it differs in measurements

The free mandatory way of improving both kinds of radiations is to make the Dot of the inductor or the start of the winding of the inductor at the Noisy switching node because of the Direction of magnetic flux generated based on Lenz law the right-hand rule as shown

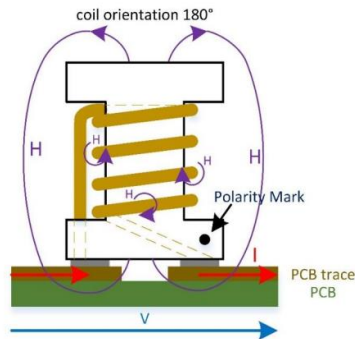


FIGURE 4: MAGNETIC FLUX AND H FIELD WITH COIL ORIENTATION OF 180°

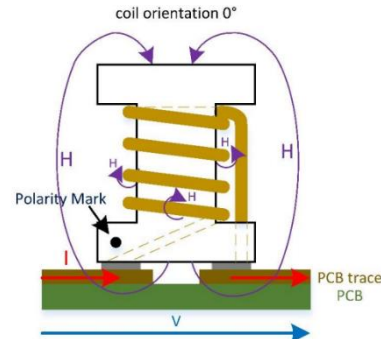
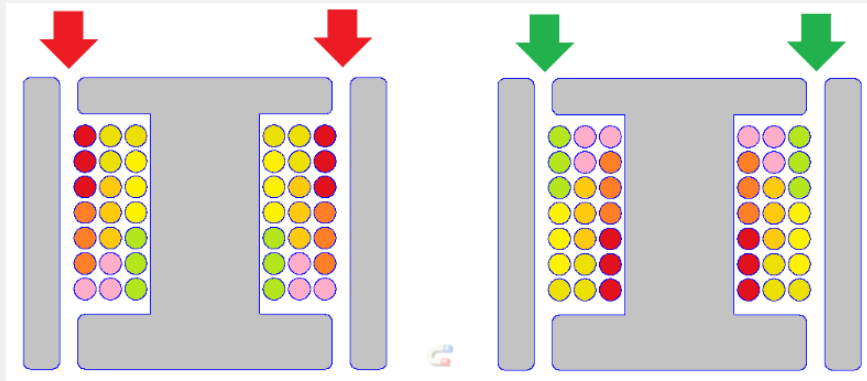


FIGURE 3: MAGNETIC FLUX AND H FIELD WITH COIL ORIENTATION OF 0°

as you the photo in the left photo that the flux is generated from top to bottom and the left photo the flux is generated from Bottom to Top.

On the other hand, if the “noisy” voltage is connected to the start of the winding (e.g. marked with a dot), so that the noisy voltage is on the *inside* of the winding, and the quiet voltage is on the outside, then the quiet outside turns act as an electrostatic shield and significantly attenuate the Radiations then the quiet outside turns act as an electrostatic shield and significantly attenuate the

Polarity of winding in a multi-layer inductor: red - “noisy” voltage, orange/yellow - medium voltage, green - “quiet” voltage



Measurement from application note regarding influences of orientation of the switching inductor in terms of Magnetic Radiation

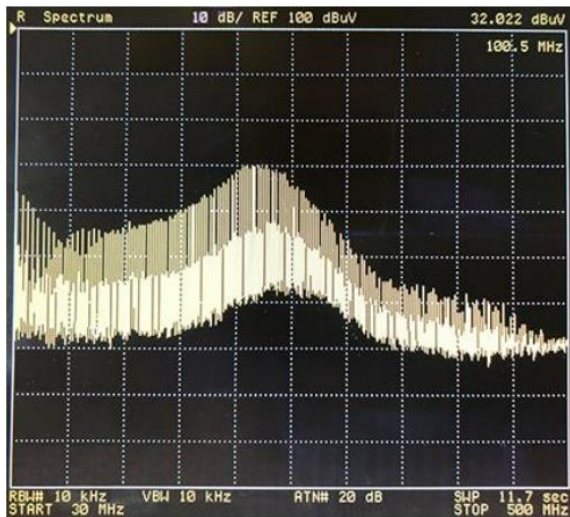


FIGURE 10: MAGNETIC FIELD AROUND THE COIL, 30 MHz TO 500 MHz, WITH 0° ROTATION

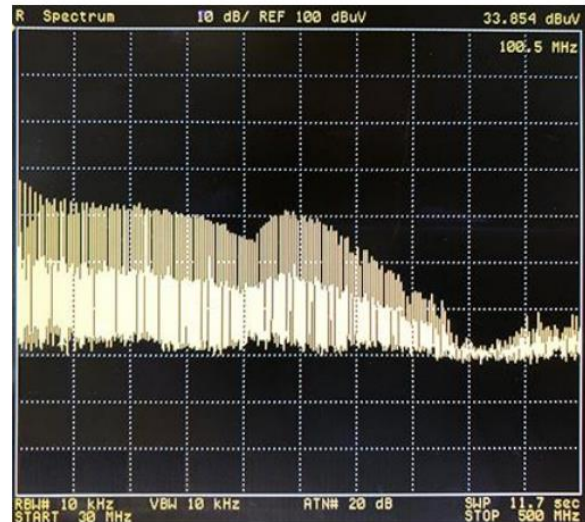


FIGURE 11: MAGNETIC FIELD AROUND THE COIL, 30 MHz TO 500 MHz, WITH 180° ROTATION

It also has an effect on the Electric-Field Radiation

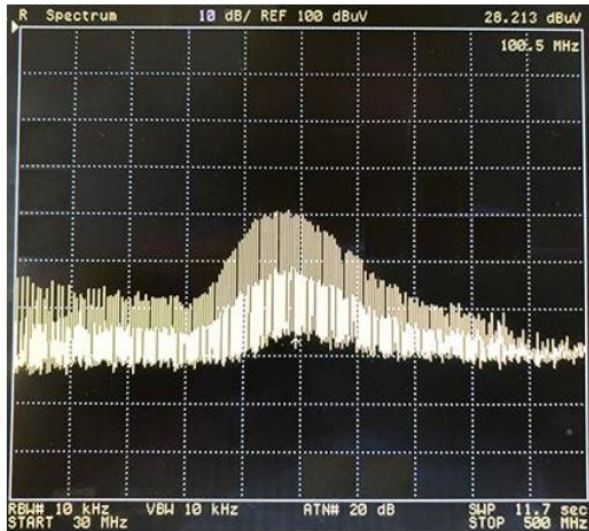


FIGURE 14: ELECTRIC FIELD AT THE TOP OF THE COIL, 30 MHz TO 500 MHz, WITH 0° ROTATION

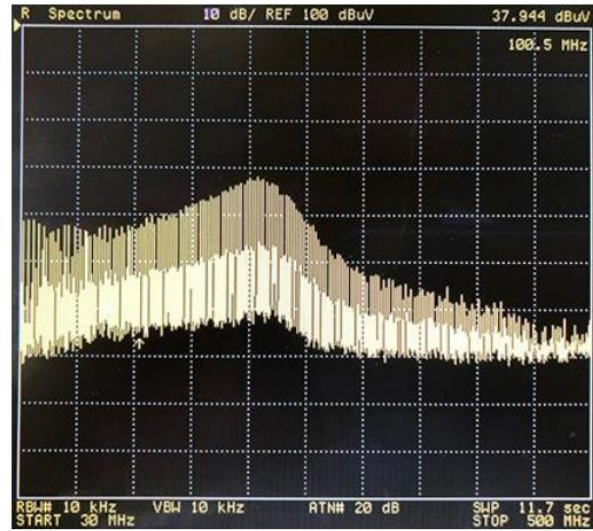


FIGURE 15: ELECTRIC FIELD AT THE TOP OF THE COIL, 30 MHz TO 500 MHz, WITH 180° ROTATION

Note the Following

The SMPS converter emits electromagnetic noise through radiation of common mode currents from cables or antennas formed in loops in the design. Radiation is complex and depends on a number of factors. These factors include the common-mode current in the cables, the differential-mode noise at the SMPS input, the cable lengths, the loops in the power traces on the SMPS PCB, high dV/dt at the switch node, the coil polarity, and the coil size.

When we say Magnetic Radiation, we mean the effect of the Normal current flowing through the inductor making Magnetic flux which is named Differential-Mode Current of the circuit mainly depending in Normal Current and Inductance of the inductor

When we say Electric-Radiation we meant the effect of the parasitic exists in the circuit including the components is self which is called Common-Mode Current

Parasitics are harder to deal with because you have unwanted effects OF R-L-C in you circuits causing oscillations hence radiations

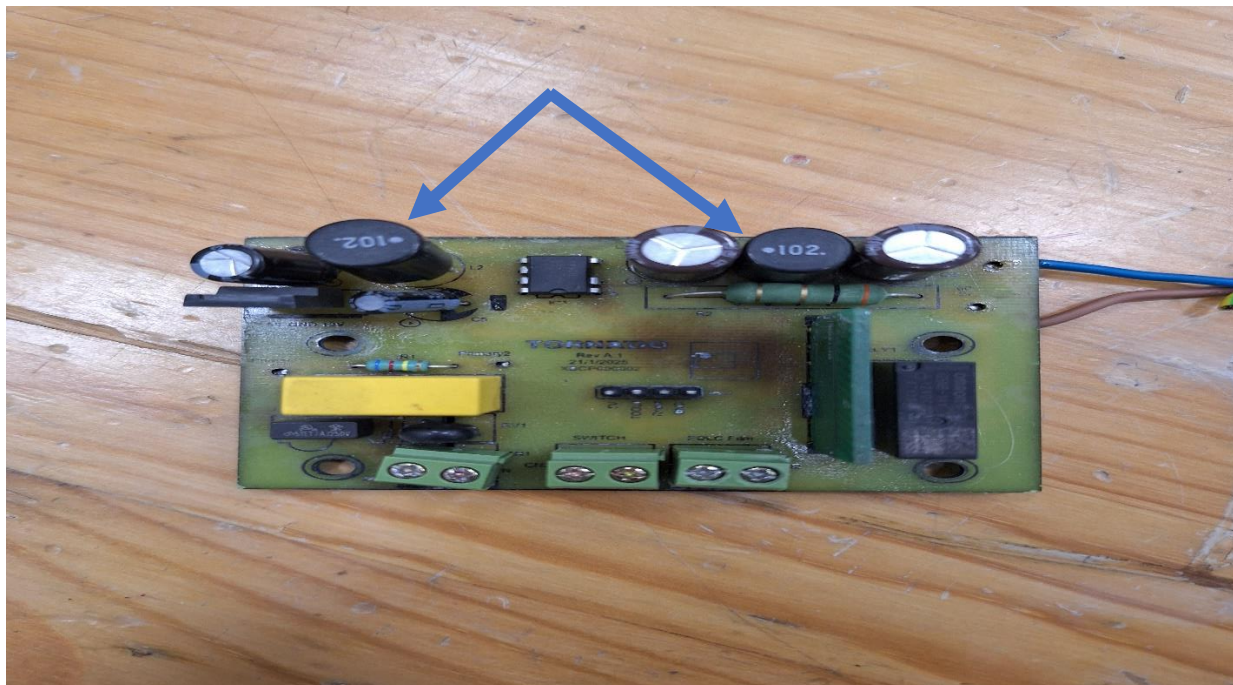
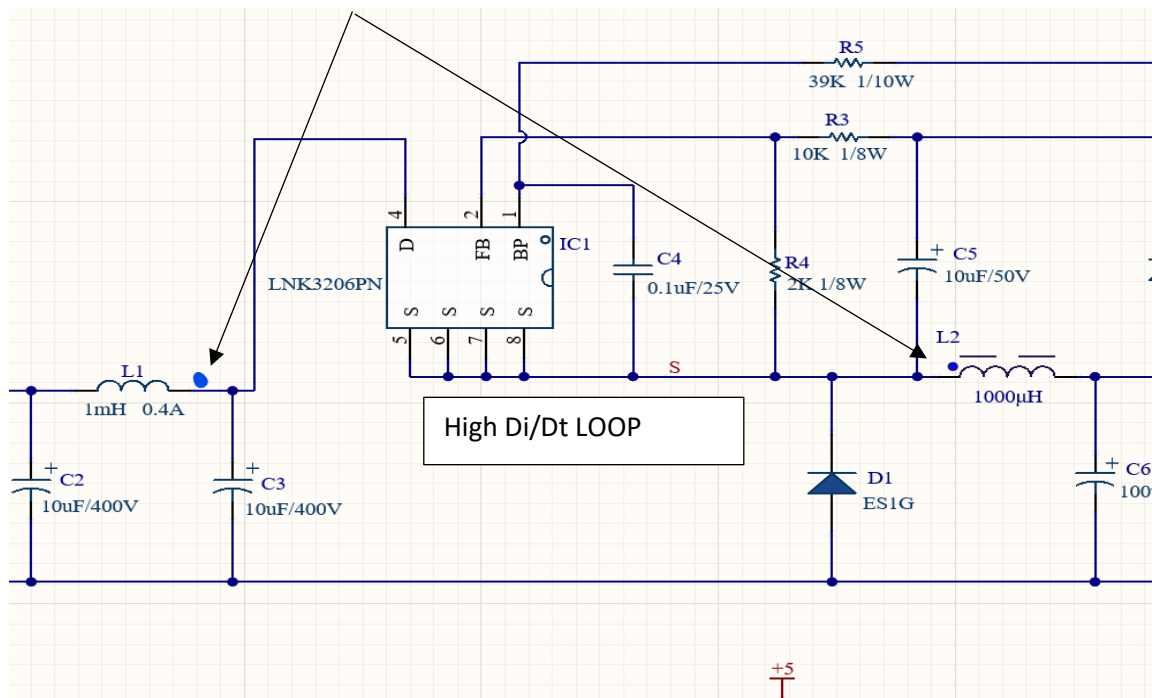
One the main issue of this common-mode behavior exists in the switching node of the converter

Where very high dV/dt A fast-switching MOSFET causes oscillations on the switch node in a frequency range of 40 MHz to 300 MHz because of parasitic effects from MOSFET and PCB traces, depending on the switching frequency of the SMPS. These oscillations create electromagnetic radiations, which may also depend on the coil polarity, over a broadband frequency range.

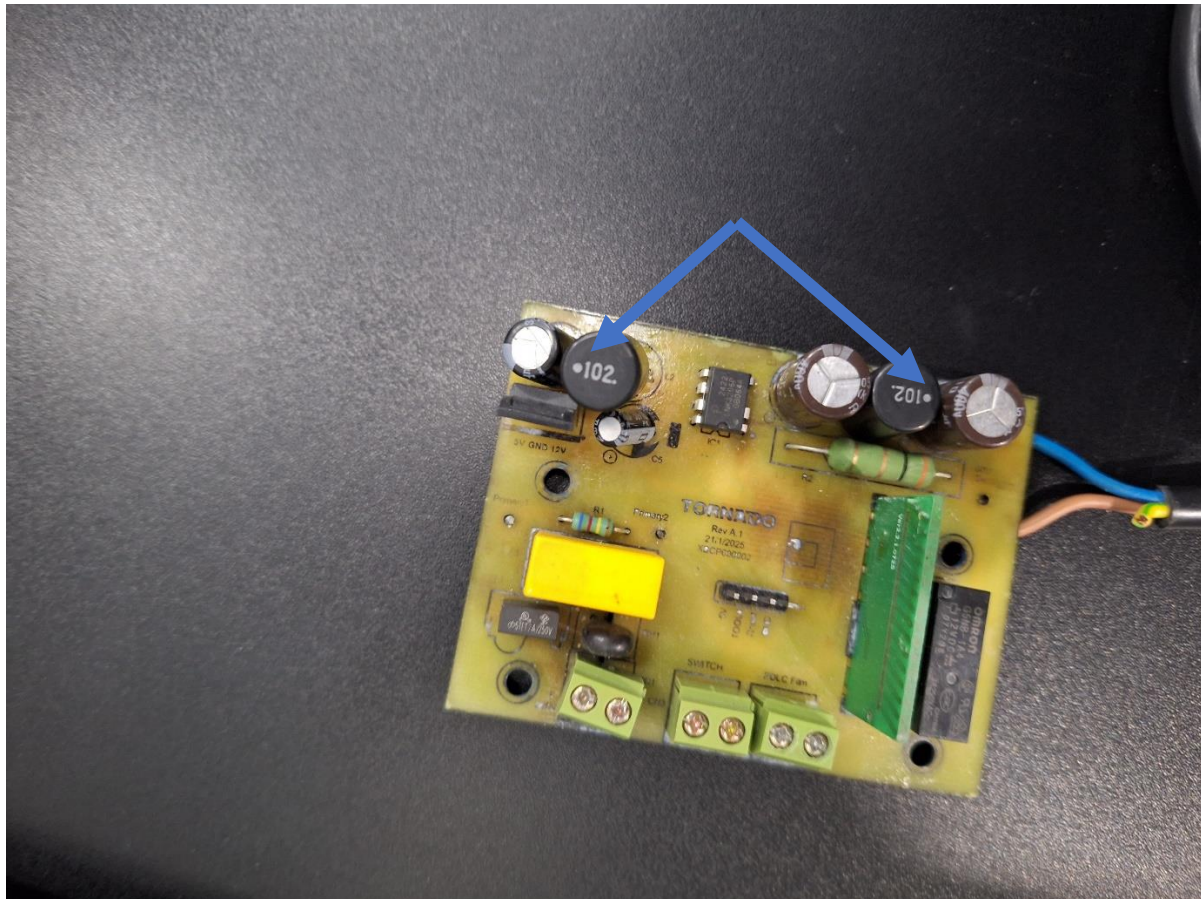
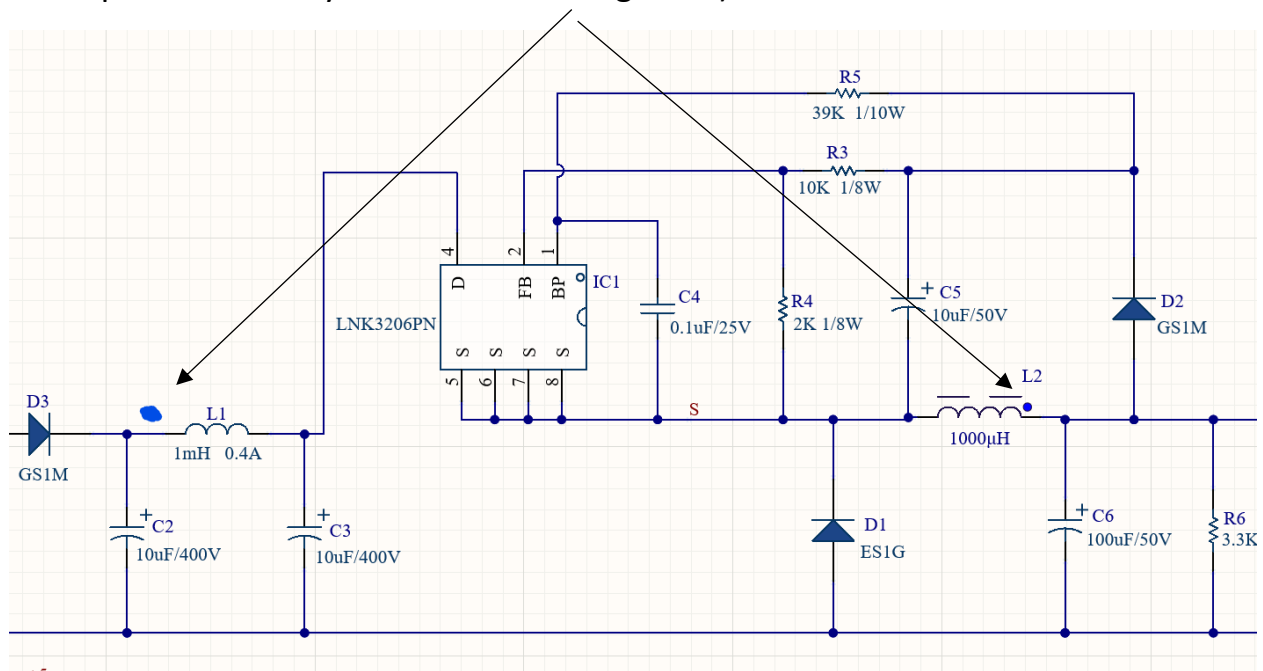
Prove of the concept theory with our lab here is a board designed earlier

The test was done with 2 trials on the same board to show the impact of the orientation only nothing else

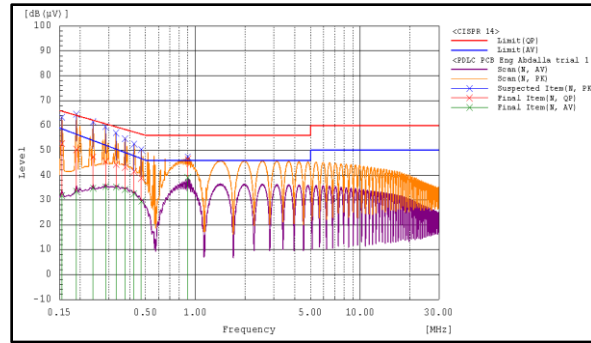
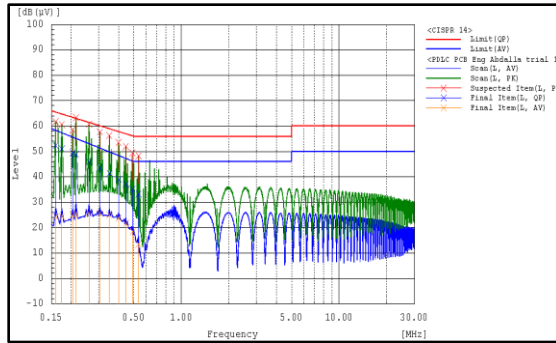
First one is the board with the right orientation of the switching inductor (the dot is placed near the switching node)



second trial is the board with the wrong orientation of the switching inductor (the dot is placed far away from the switching node)

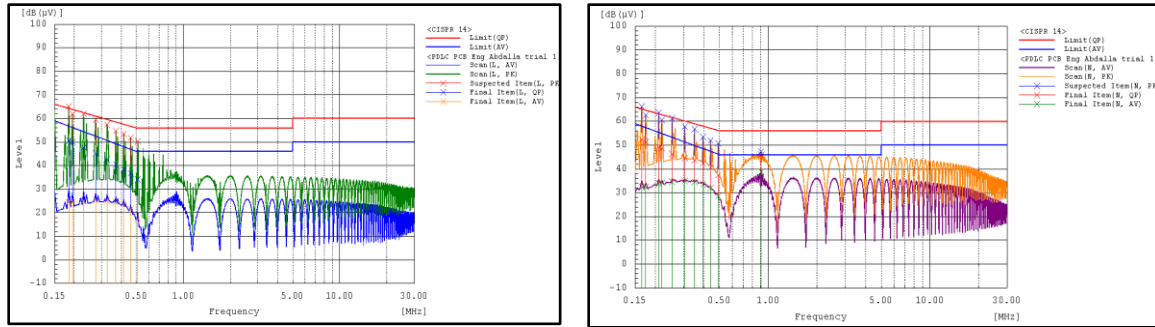


Trial 1



Frequency [MHz]	Line	Reading QP [dB(μV)]	Reading AV [dB(μV)]	Factor [dB]	Level QP [dB(μV)]	Level AV [dB(μV)]	Limit QP [dB(μV)]	Limit AV [dB(μV)]	Margin QP [dB]	Margin AV [dB]
0.160	L	42.2	14.2	10.2	52.4	24.4	65.5	58.3	13.1	33.9
0.175	L	40.9	14.2	10.2	51.1	24.4	64.7	57.3	13.6	32.9
0.205	L	39.4	14.9	10.2	49.6	25.1	63.4	55.6	13.8	30.5
0.215	L	38.7	15.2	10.2	48.9	25.4	63.0	55.1	14.1	29.7
0.260	L	35.9	15.6	10.2	46.1	25.8	61.4	53.1	15.3	27.3
0.305	L	33.4	15.5	10.2	43.6	25.7	60.1	51.3	16.5	25.6
0.350	L	31.2	14.9	10.2	41.4	25.1	59.0	49.9	17.6	24.8
0.400	L	28.8	13.7	10.2	39.0	23.9	57.9	48.4	18.9	24.5
0.445	L	27.0	11.5	10.2	37.2	21.7	57.0	47.3	19.8	25.6
0.490	L	25.0	8.3	10.2	35.2	18.5	56.2	46.2	21.0	27.7
0.535	L	23.1	3.3	10.2	33.3	13.5	56.0	46.0	22.7	32.5
0.155	N	42.5	21.5	10.2	52.7	31.7	65.7	58.6	13.0	26.9
0.190	N	40.2	23.3	10.2	50.4	33.5	64.0	56.4	13.6	22.9
0.240	N	37.2	24.6	10.2	47.4	34.8	62.1	53.9	14.7	19.1
0.285	N	35.4	25.2	10.2	45.6	35.4	60.7	52.1	15.1	16.7
0.330	N	34.2	25.0	10.2	44.4	35.2	59.5	50.5	15.1	15.3
0.375	N	33.0	24.1	10.2	43.2	34.3	58.4	49.1	15.2	14.8
0.425	N	31.2	22.4	10.2	41.4	32.6	57.3	47.8	15.9	15.2
0.470	N	28.5	19.3	10.2	38.7	29.5	56.5	46.7	17.8	17.2
0.900	N	36.2	28.9	10.2	46.4	39.1	56.0	46.0	9.6	6.9

Trial 2



Frequency [MHz]	Line	Reading QP [dB(μV)]	Reading AV [dB(μV)]	Factor [dB]	Level QP [dB(μV)]	Level AV [dB(μV)]	Limit QP [dB(μV)]	Limit AV [dB(μV)]	Margin QP [dB]	Margin AV [dB]
0.185	L	40.5	14.3	10.2	50.7	24.5	64.3	56.7	13.6	32.2
0.195	L	39.8	14.5	10.2	50.0	24.7	63.8	56.2	13.8	31.5
0.230	L	37.5	14.8	10.2	47.7	25.0	62.4	54.4	14.7	29.4
0.275	L	35.0	15.1	10.2	45.2	25.3	61.0	52.5	15.8	27.2
0.325	L	32.5	15.0	10.2	42.7	25.2	59.6	50.7	16.9	25.5
0.370	L	30.2	14.4	10.2	40.4	24.6	58.5	49.3	18.1	24.7
0.415	L	28.2	13.3	10.2	38.4	23.5	57.5	48.0	19.1	24.5
0.460	L	26.6	11.4	10.2	36.8	21.6	56.7	46.9	19.9	25.3
0.510	L	24.0	7.4	10.2	34.2	17.6	56.0	46.0	21.8	28.4
0.165	N	42.2	22.2	10.2	52.4	32.4	65.2	58.0	12.8	25.6
0.175	N	41.2	22.2	10.2	51.4	32.4	64.7	57.3	13.3	24.9
0.210	N	39.0	23.7	10.2	49.2	33.9	63.2	55.4	14.0	21.5
0.220	N	38.5	23.9	10.2	48.7	34.1	62.8	54.9	14.1	20.8
0.255	N	36.2	24.4	10.2	46.4	34.6	61.6	53.3	15.2	18.7
0.305	N	34.4	24.7	10.2	44.6	34.9	60.1	51.3	15.5	16.4
0.350	N	33.5	24.3	10.2	43.7	34.5	59.0	49.9	15.3	15.4
0.395	N	32.3	23.4	10.2	42.5	33.6	58.0	48.5	15.5	14.9
0.440	N	30.5	21.9	10.2	40.7	32.1	57.1	47.4	16.4	15.3
0.490	N	27.1	18.3	10.2	37.3	28.5	56.2	46.2	18.9	17.7
0.905	N	35.8	28.6	10.2	46.0	38.8	56.0	46.0	10.0	7.2

The improvements appeared at Frequency of 900 MHz and it was about 0.4 dB free of charge improvement

Notice that its very logic to start seeing the improvement at near 1 MHz

Due to the radiation Dominant effect accrue at very high frequency

And THE IMPROVEMENT IN THE DV test was about 0.3 to 0.4 dB improvements and that is free of charge improvement Typically from 30 MHz to 500 MHz or higher (Electric-Field and magnetic Field Radiation)